

* L:8 Pro. for ml

⇒ Revision:

① Adj R^2 :

$$1 - \frac{(1 - R^2)(n-1)}{(n-1-k)}$$

n = no. of datapoints

k = no. of features

When we use?

to compare model
with diff. no. of
featu.

→ diff between R^2 & Adj R^2 :

- R^2 ↑ even if we add unnec. features?

Why? → overfitting. (Not R^2 → training R^2)

- Adj R^2 pen. unnec. features.

→ Limitation: It only useful with L.R. models.
(SLR, MLR & Poly)

Why?

→ bcz In other non linear model we can't
capture comp. using only k (no. of fea).

→ which is more complex model?

$$y = mx + b$$

$$y = m_1x_1 + m_2x_2 + m_3x_3 + m_4x_4 + c$$

* Set: Collection of unique (distinct) elements.

Ex: $\{1, 2, 3, 4, 5, 6\}$ $\{1, 2, 3\} = \{3, 2, 1\}$

→ NO Repetition, NO order.

* Ops on set:

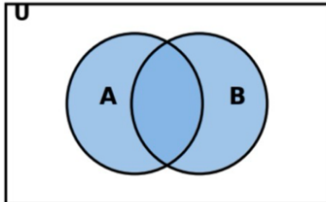
Symbol	Read as	Meaning	Example
$\in$	is an element of	The object belongs to the set	$3 \in \{1, 2, 3\}$
$\notin$	is not an element of	The object does not belong to the set	$5 \notin \{1, 2, 3\}$
$\subseteq$	is a subset of	Every element of the first set is also in the second	$\{1, 2\} \subseteq \{1, 2, 3\}$
U	universal set	The set of everything under discussion in the problem	$U = \{1, 2, 3, 4, 5, 6\}$ for a die
$\emptyset$ or $\{\}$	empty set	A set with no elements at all	$\{x : x \text{ is a die face greater than } 6\} = \emptyset$
$n(A)$	size of A	The number of elements in set A	$n(\{2, 4, 6\}) = 3$
$\cup$	union	Everything in A or in B or in both	$\{1, 2\} \cup \{2, 3\} = \{1, 2, 3\}$
$\cap$	intersection	Only what is in A and in B at the same time	$\{1, 2\} \cap \{2, 3\} = \{2\}$
A' or A^c	complement of A	Everything in U that is not in A	If $U = \{1..6\}$, $A = \{2, 4, 6\}$ then $A' = \{1, 3, 5\}$
$A - B$	A minus B	In A but not in B	$\{1, 2, 3\} - \{2\} = \{1, 3\}$

Two sets with nothing in common are called disjoint: $A \cap B = \emptyset$.

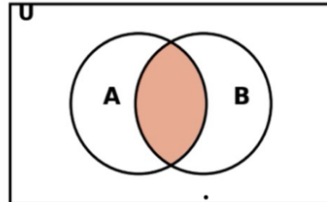
* Venn diagram:

A **Venn diagram** draws the universal set as a rectangle and each set as a circle inside it. Overlapping circles share elements.

$A \cup B$ (Union)



$A \cap B$ (Intersection)



A' (Complement of A)

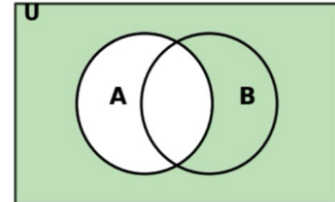


Figure 1.1 — Union, intersection and complement shown on Venn diagrams.

Ex: ①

$$A = \{3, 4, 5\}$$

$$B = \{1, 2, 3\}$$

$$n(A) = (?)$$

$$n(B) = (?)$$

$$n(A \cup B) = (?)$$

$$A \cup B = \{1, \dots, 5\}$$

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

②

In a class of 40 students, 22 study Python, 18 study SQL and 9 study both.

To find: How many study at least one of the two languages, and how many study neither
?

$$n(U): 40 \quad n(P) = 22 \quad n(S) = 18$$

$$n(P \cap S) = 9$$

$$\begin{aligned} n(P \cup S) &= n(P) + n(S) - n(P \cap S) \\ &= 22 + 18 - 9 \end{aligned}$$

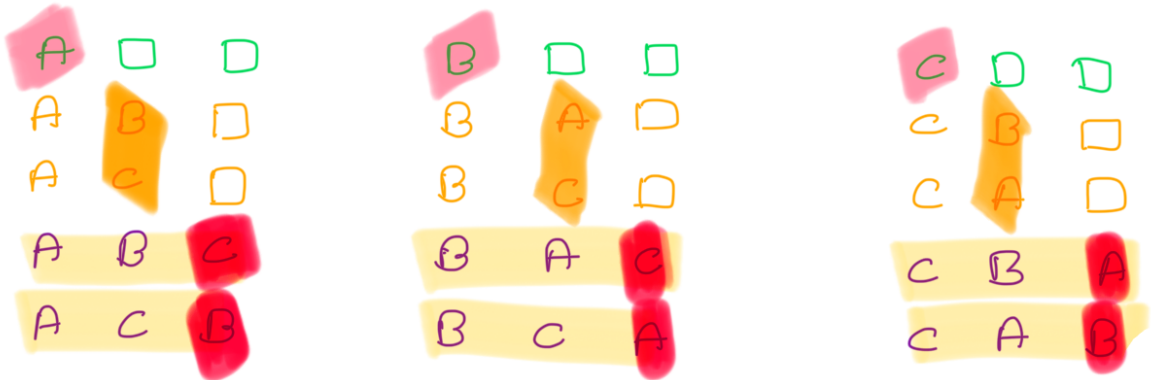
$$= 3!$$

$$n(\text{pus})' = 9$$

* Counting techniques:

Set: $\{A, B, C\}$

$\square \square \square \leftarrow$ How many diff. arrangements are possible?



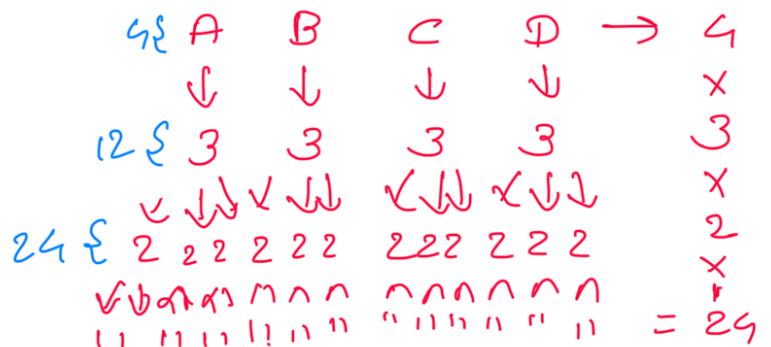
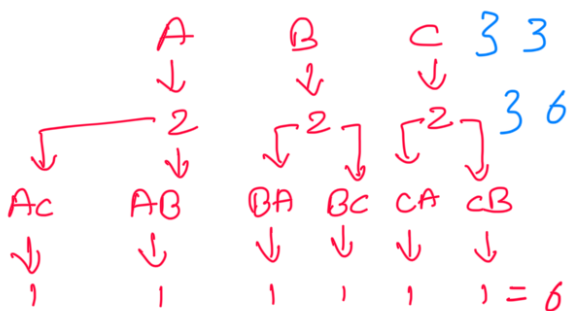
⑥

$$\begin{array}{ccc} D & D & D \\ \uparrow & \uparrow & \uparrow \\ 3 & \times & 2 \times 1 = 6 \end{array}$$

$\rightarrow 3!$

First pos: 3 choice

Not for each chosen one have 2 choice



* $n!$ = No. of ways to arrange n distinct objects is $n!$.

$$3 \rightarrow 3!$$

$$4 \rightarrow 4!$$

$$5 \rightarrow 5!$$

$$0! \rightarrow 1$$

How many ways can we arrange zero objects?

→ one : Do nothing

$$5! = 5 \times 4 \times 3 \times 2 \times 1$$

$$4! = 4 \times 3 \times 2 \times 1$$

$$5! = 5 \times 4!$$

$$4! = 4 \times 3! \rightarrow 3! = \frac{4!}{4}$$

$$3! = 3 \times 2!$$

$$2! = 2 \times 1!$$

$$1! = 1 \times 0!$$

$$2! = \frac{3!}{3}$$

$$1! = \frac{2!}{2}$$

$$0! = \frac{1!}{1} = 1$$

* n_{P_r} :

Set: $\{A, B, C\}$

□ □

⑥

AD

AB

AC

BD

BA

BC

CD

CA

CB

$$\{A, B, C, \emptyset\} \rightarrow 4! = 4 \times 3 \times 2 \times 1 = \underline{\underline{24}}$$

□ □

AD BD CD $\emptyset$ □
 AB BC C $\emptyset$ $\emptyset$ A
 AC B $\emptyset$ CA $\emptyset$ B
 A $\emptyset$ BA CB $\emptyset$ C

} (12)

$$\frac{n!}{2}$$

$$\frac{24}{2} = (12)$$

→ nPr : The no. of ways to arrange r objects chosen from (n) distinct object.

↓
4

$$nPr = \frac{n!}{(n-r)!}$$

A B C D E

D D D D D

$$\begin{array}{ccccccccc} A & B & C & D & E & & = & 5 \\ 1 & 1 & 1 & 1 & 1 & & : & 5 \\ 4 & 4 & 4 & 4 & 4 & & : & (20) \end{array}$$

$$\begin{array}{ccccccc} 1 & 2 & 3 & \dots & 20 \\ \downarrow & \downarrow & \downarrow & & \downarrow \\ 3 & 3 & 3 & \dots & 3 & : & 60 \end{array}$$

$$\begin{array}{ccccccc} 1 & \dots & 60 \\ 1 & 1 & 1 & & 1 \\ 2 & 2 & 2 & \dots & 2 & : & 120 \end{array}$$

$$1 \dots 120$$

$$1 \cdot \dots \cdot 120$$

$$1 \cdot \dots \cdot 1$$

$$1 \cdot \dots \cdot 1 : 120$$

$$5! = 5 \times 4 \times \boxed{3 \times 2 \times 1}$$

$\underbrace{\hspace{10em}}_{20}$
 $\underbrace{\hspace{10em}}_{60}$
 $\underbrace{\hspace{10em}}_{120}$
 $\underbrace{\hspace{10em}}_{120}$

$3!$